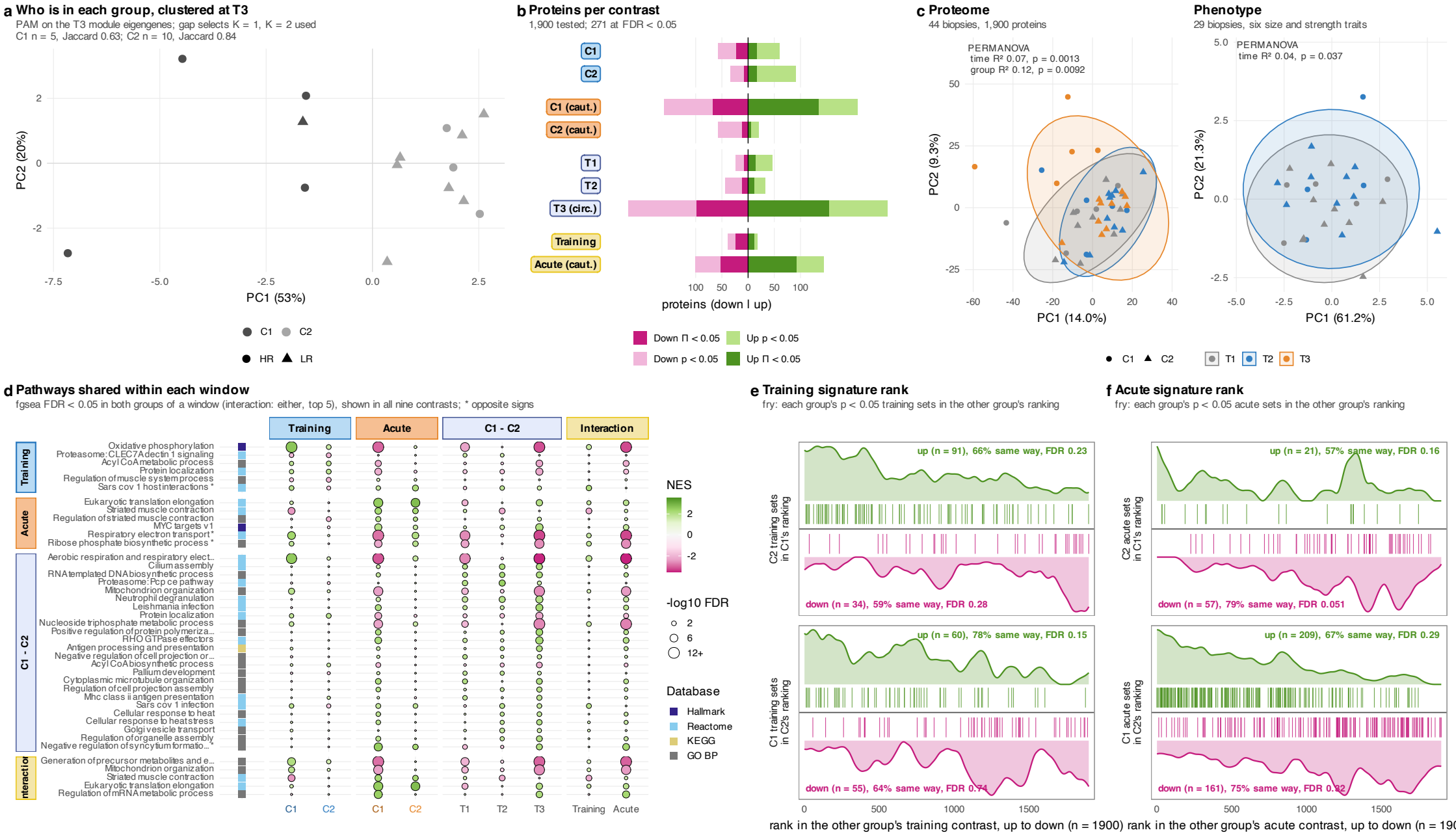

a Who is in each group, clustered at T3
PAM on the T3 module eigengenes; gap selects K = 1, K = 2 used
C1 n = 5, Jaccard 0.63; C2 n = 10, Jaccard 0.84
PC2 (20%)
PC1 (53%)
C1 C2
HR LR
b Proteins per contrast
1,900 tested; 271 at FDR < 0.05
C1
C2
C1 (caut.)
C2 (caut.)
T1
T2
T3 (circ.)
Training
Acute (caut.)
proteins (down | up)
Down Π < 0.05
Up p < 0.05
Down p < 0.05
Up Π < 0.05
c Proteome
44 biopsies, 1,900 proteins
PERMANOVA
time R² 0.07, p = 0.0013
group R² 0.12, p = 0.0092
PC2 (9.3%)
PC1 (14.0%)
Phenotype
29 biopsies, six size and strength traits
PERMANOVA
time R² 0.04, p = 0.037
PC2 (21.3%)
PC1 (61.2%)
C1 C2
T1 T2 T3
d Pathways shared within each window
fgsea FDR < 0.05 in both groups of a window (interaction: either, top 5), shown in all nine contrasts; * opposite signs
Training
Acute
C1 - C2
Interaction
Oxidative phosphorylation
Proteasome:CLEC7Adectin 1 signaling
Acyl CoA metabolic process
Protein localization
Regulation of muscle system process
Sars cov 1 host interactions *
Eukaryotic translation elongation
Striated muscle contraction
Regulation of striated muscle contraction
MYC targets v1
Respiratory electron transport *
Ribose phosphate biosynthetic process *
Aerobic respiration and respiratory elect...
Cilium assembly
RNA templated DNA biosynthetic process
Proteasome:Pcp ce pathway
Mitochondrion organization
Neutrophil degranulation
Leishmania infection
Protein localization
Nucleoside triphosphate metabolic process
Positive regulation of protein polymeriza...
RHO GTPase effectors
Antigen processing and presentation
Negative regulation of cell projection or...
Acyl CoA biosynthetic process
Pallium development
Cytoplasmic microtubule organization
Regulation of cell projection assembly
Mhc class ii antigen presentation
Sars cov 1 infection
Cellular response to heat
Cellular response to heat stress
Golgi vesicle transport
Regulation of organelle assembly
Negative regulation of syncytium formatio... *
Generation of precursor metabolites and e...
Mitochondrion organization
Striated muscle contraction
Eukaryotic translation elongation
Regulation of mRNA metabolic process
C1 C2 C1 C2 T1 T2 T3 Training Acute
NES
-log10 FDR
Database
Hallmark
Reactome
KEGG
GO BP
e Training signature rank
fry: each group's p < 0.05 training sets in the other group's ranking
C2 training sets in C1's ranking
up (n = 91), 66% same way, FDR 0.23
down (n = 34), 59% same way, FDR 0.28
C1 training sets in C2s ranking
up (n = 60), 78% same way, FDR 0.15
down (n = 55), 64% same way, FDR 0.74
rank in the other group's training contrast, up to down (n = 1900)
f Acute signature rank
fry: each group's p < 0.05 acute sets in the other group's ranking
C2 acute sets in C1's ranking
up (n = 21), 57% same way, FDR 0.16
down (n = 57), 79% same way, FDR 0.051
C1 acute sets in C2s ranking
up (n = 209), 67% same way, FDR 0.29
down (n = 161), 75% same way, FDR 0.92
rank in the other group's acute contrast, up to down (n = 1900)